* Practical - 20 *

* activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="16dp"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/locationText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Fetching location..."
        android:textSize="18sp"
        android:padding="16dp"
        android:textStyle="bold" />

    <Button
        android:id="@+id/btnGetLocation"
        android:text="Get Location"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:padding="12dp"
        android:layout_marginTop="20dp" />
</LinearLayout>
```

* MainActivity.java

```
package com.example.locationapp;

public class MainActivity extends AppCompatActivity {

    private static final int REQUEST_CODE = 101;
    private FusedLocationProviderClient fusedLocationClient;
    private TextView locationText;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        locationText = findViewById(R.id.locationText);
        fusedLocationClient = LocationServices.
                        getFusedLocationProviderClient(this);

        getLastLocation();
    }

    private void getLastLocation() {
        if (ActivityCompat.checkSelfPermission
    (this, Manifest.permission.ACCESS_FINE_LOCATION)
    != PackageManager.PERMISSION_GRANTED) {
        ActivityCompat.requestPermissions(this, new String[]
    {Manifest.permission.ACCESS_FINE_LOCATION},
    REQUEST_CODE);
    return;
  }
```

```
fusedLocationClient.getLastLocation().addOnSuccessListener
    (this, new OnSuccessListener<Location>() {
    @Override
    public void onSuccess(Location location) {
        if (location != null) {
            double lat = location.getLatitude();
            double lon = location.getLongitude();

            String locText = getString(R.string.
            location_text, lat, lon);

        Geocoder geocoder = new Geocoder(MainActivity.
        this, Locale.getDefault());
        try{
            List<Address> addresses = geocoder.getFromLocation
            (lat, lon, 1);
            if(address != null && !addresses.isEmpty()){
                Address address = addresses.get(0);
                String fullAddress = address.getAddressLine(0);
                locText += "\nAddress: " + fullAddress;
            }
        }catch (IOException e) {
            e.printStackTrace();
            Toast.makeText(MainActivity.this, "Unable to
            get address", Toast.LENGTH_SHORT).show();
        }

        locationText.setText(locText);

    }else{
    Toast.makeText(MainActivity.this, getString(R.string.
```

```
        location_not_fount), Toast.LENGTH_SHORT).show();
      }
    }
  });
}

@Override
public void onRequestPermmissionsResult (int requestCode,
@NonNull String[] permissions, @NonNull int[] grantResults){
    super.onRequestPermissionsResult (requestCode,
    permissions, grantResults);
    if (requestCode == REQUEST_CODE &&
        grantResults.length > 0 && grantResults[0]
        == PackageManager.PERMISSION_GRANTED){
      getlastlocation();
    }else {
      Toast.makeText(this, getString(R.string.
      permission_denied), Toast.LENGTH_SHORT).show();
    }
  }
}
```